

30-Day React Internals Roadmap

WEEK 1 — Hooks Internals (Foundation)

- Day 1–2: Study `ReactHooks.js` and understand `resolveDispatcher()`.
- Day 3–4: Deep dive into `ReactFiberHooks.js` (`mountState`, `updateState`, `dispatchSetState`).
- Day 5–6: Build a tiny custom `useState` implementation in plain JavaScript.
- Day 7: Explain why hooks must not be conditional and why order matters.

WEEK 2 — Fiber Architecture

- Day 8–9: Study `ReactFiber.js` and understand Fiber node structure.
- Day 10–11: Learn `beginWork` and `completeWork` (render phase).
- Day 12–13: Understand commit phase and `commitRoot`.
- Day 14: Draw the complete flow from component render to commit.

WEEK 3 — Scheduler & Updates

- Day 15–16: Study update queues inside `ReactFiberHooks.js`.
- Day 17–18: Explore the scheduler package and concurrent rendering basics.
- Day 19–20: Learn how `StrictMode` works and why double rendering happens in DEV.
- Day 21: Debug React internals using breakpoints.

WEEK 4 — Mastery Through Application

- Day 22–23: Build a mini React implementation (`createElement`, `render`, `useState`).
- Day 24–25: Read React RFCs to understand design decisions.
- Day 26–27: Deep dive into reconciliation and keys.
- Day 28: Study performance tools (`memo`, `useMemo`, `useCallback`).
- Day 29: Teach the full architecture out loud.
- Day 30: Refactor a real project using your new mental model.

